

Comparison of MPC, Task-Space PID and Time-delayed Control

Stage Report – 3-DOF OpenManipulator-X

Jiecheng Zhu

Supervisor: Prof. Dr. Ping Zhang

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What was compared

Three controllers holding a task-space reference on the same arm:

- MPC with a Kalman disturbance observer (Cao & Tang)
- Task-space PID – the baseline
- Time-delayed Control in its basic joint-space form (Memmer)

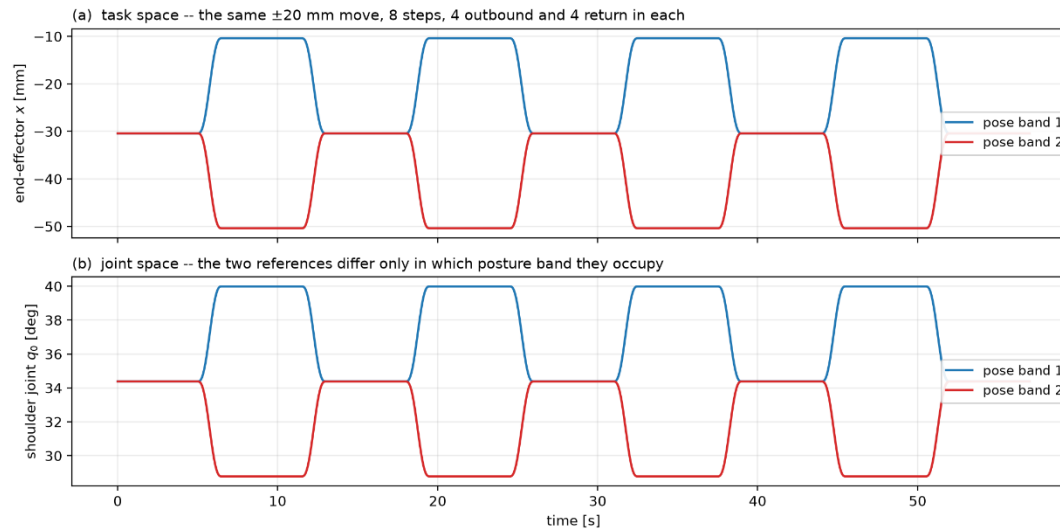
Experiment design

All three run the **same step-response experiment**: one reference trajectory, one loop rate (100 Hz), two arm postures and one set of metric definitions — described on the next slide.

Every headline number is **3 runs × 8 steps** = 24 steps.

One trajectory, two postures

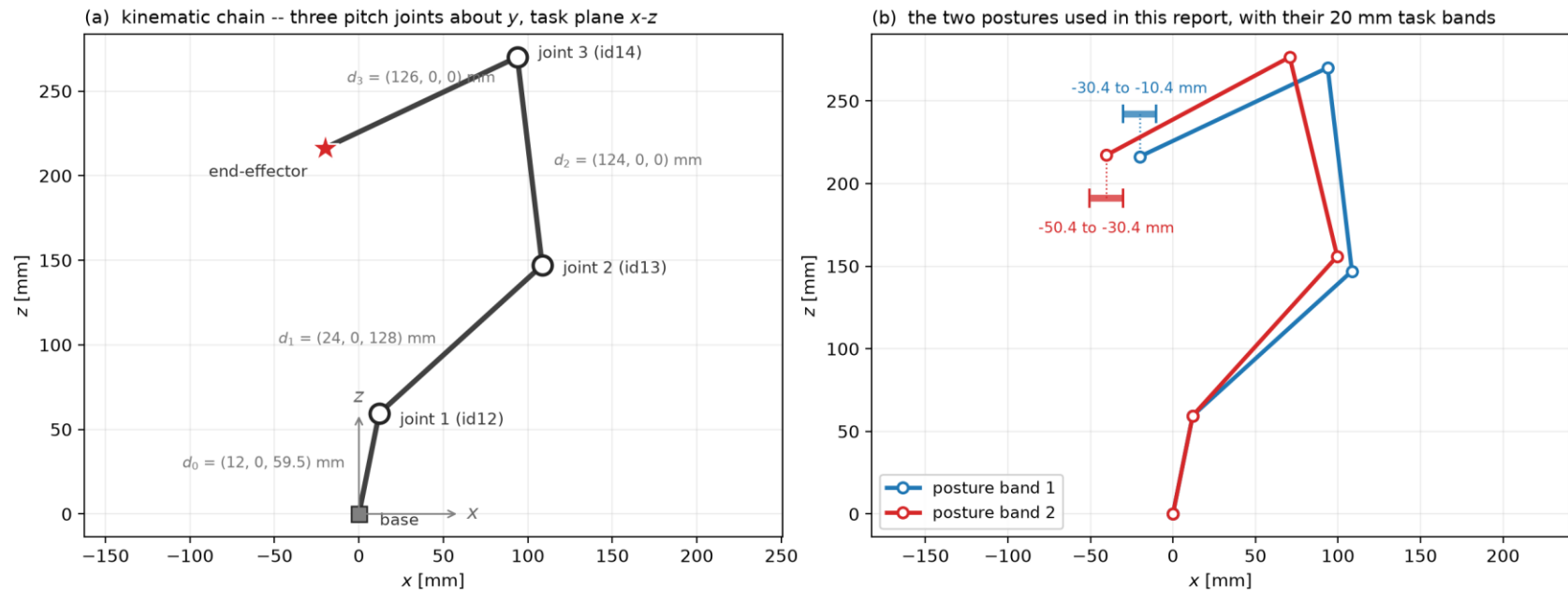
Fig. 4.2 The step reference used throughout, in both pose bands



- One run = hold the tool still, command a 20 mm step along x , wait 6.5 s, step back. 8 steps, 57 s.
- Two posture bands = the same 20 mm move on either side of one start pose. Each has **4 outbound and 4 return** moves, so the variable is the arm configuration, not the direction.
- Scored per step: rise (10–90 %), overshoot, settling (last exit from ± 3 mm), settled $|e_x|$ (last 2 s of the dwell).

Manipulator and test trajectory

Fig. 2.1 The 3-DOF waist-removed OpenManipulator-X



pitch joints, waist removed · Current Control Mode · task plane x-z · 10 ms tick

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \boldsymbol{\tau} + \mathbf{J}^T(\mathbf{q})\mathbf{F}_h$$

$\mathbf{M}(\mathbf{q})$ is the (positive-definite) joint-space inertia matrix

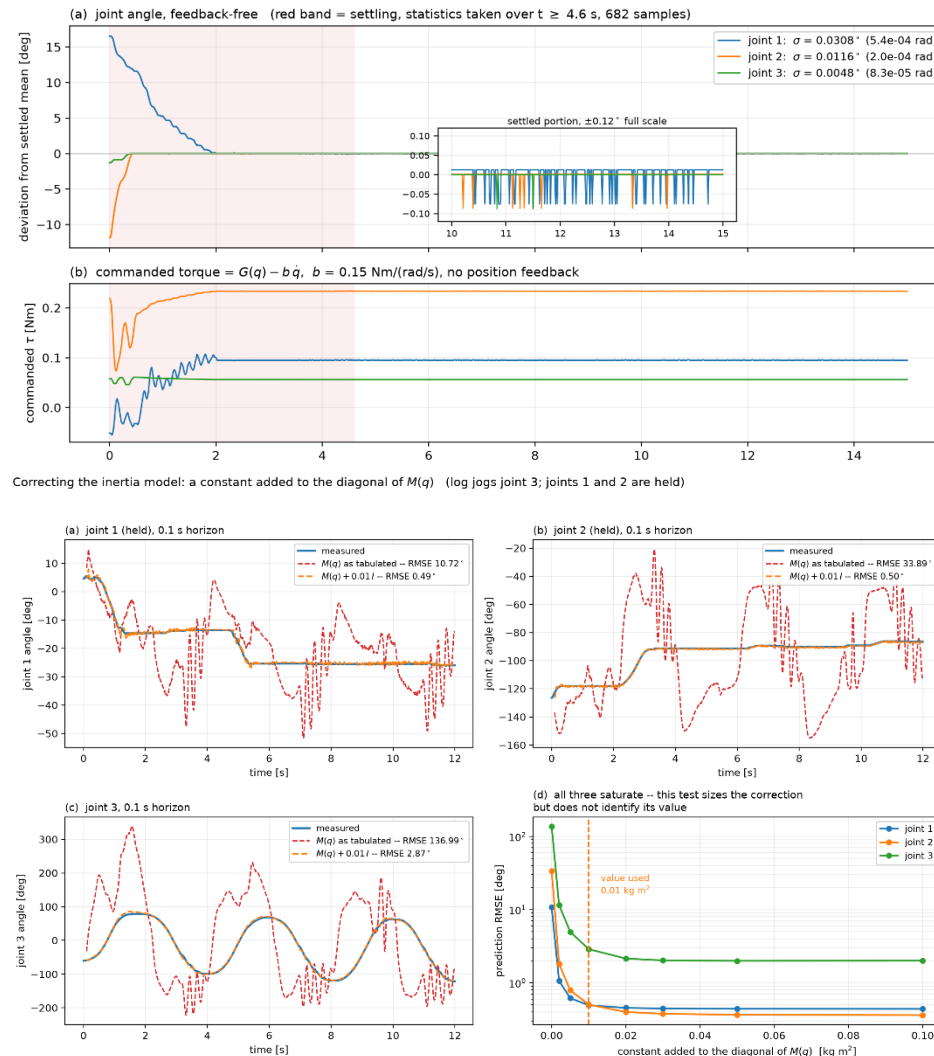
$\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ is the Coriolis/centrifugal matrix

$\mathbf{G}(\mathbf{q})$ is the gravity torque

$\mathbf{F}_h$ is any external (e.g. human-contact) wrench acting through the full geometric Jacobian $\mathbf{J}(\mathbf{q})$

The manipulator tested here is 3-DoF and the task is purely translational and 2-D (x, z only)

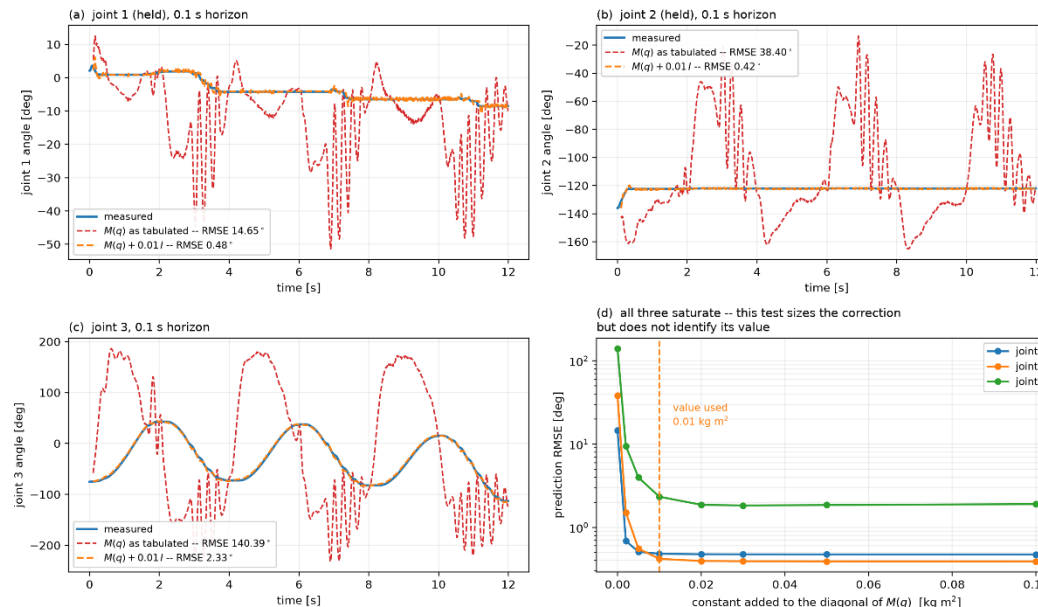
Fig. 2.2 Static validation of the gravity model: all three joints hold within encoder noise



- Statically the gravity model is right: $\sigma \leq 0.031^\circ$ with no feedback
- In open-loop test all three joints diverge within 0.1 s (10 steps), worst on joint 3, improved by adjust the inertia diagonal.
- Result: error $137.0^\circ \rightarrow 2.87^\circ$.

Fixing it costs one constant, but the test cannot size it

Fig. 2.4 Correcting the inertia model: a constant added to the diagonal of $M(q)$ (log jogs joint 3; joints 1 and 2 are held)



$M(q) + 0.01 I$: joint 3 RMSE $140.4^\circ \rightarrow 2.33^\circ$.

The open-loop test proves a correction is needed but does not identify its value. The value used was fixed by closed-loop stability.

1. **Linearise.** $\tau = J_{xz}(q)^T F + \tau_{base}$ cancels gravity and Coriolis, leaving a double integrator in task space:

$$\ddot{e} = \Lambda(q)^{-1} F + d, e = p - p_d$$

2. **Predict** $N = 10$ steps ahead: $X = \Phi x_0 + \Gamma V$, with $x_0 = [e; \dot{e}]$ measured in real time.

3. **Choose the force sequence V by minimising**

$$J = \sum_{k=0}^{N-2} q_{pos} \|e_k\|^2 + q_{vel} \|\dot{e}_k\|^2 + \sum_{k=0}^{N-1} r \|F_k\|^2$$

4. **Apply the first force only**, re-solve next tick. A Kalman observer estimates $d(t)$.

Task-space PID

$$\mathbf{F} = -K_p^{PID} \mathbf{e} - K_i^{PID} \int \mathbf{e} dt - K_d^{PID} \dot{\mathbf{e}}$$

Same $\boldsymbol{\tau} = \mathbf{J}_{xz}(\mathbf{q})^T \mathbf{F} + \boldsymbol{\tau}_{base}$ mapping as the MPC, byte-identical robot block.

TDC — joint space

$$\boldsymbol{\tau}(t) = \bar{\mathbf{D}}(K_p^{TDC} \mathbf{e}_q + K_d^{TDC} \dot{\mathbf{e}}_q + \ddot{\mathbf{q}}_d) + \boldsymbol{\tau}(t - T) - \bar{\mathbf{D}}\ddot{\mathbf{q}}(t - T), \mathbf{e}_q = \mathbf{q}_d - \mathbf{q}$$

Note the joint-space error $\mathbf{e}_q$, against the task-space $\mathbf{e}$ above. $\bar{\mathbf{D}}$ is a constant diagonal nominal inertia; it multiplies the whole bracket, so it scales loop gain directly.

So the three are tuned in three different currencies:

$$(q_{pos}, q_{vel}, r), (K_p^{PID}, K_i^{PID}, K_d^{PID}) \text{ and } (\bar{\mathbf{D}}, K_p^{TDC}, K_d^{TDC}).$$

The bridge: turning q_{pos} into N/m

Minimising J over V with no constraint active has a closed-form solution that is linear in the measured error:

$$V^* = - \underbrace{(\Gamma^T \bar{Q} \Gamma + \bar{R})^{-1} \Gamma^T \bar{Q} \Phi}_{\text{one linear solve}} x_0 \quad \Rightarrow \quad F_0 = -K(\Lambda(q)) [e; \dot{e}] - \Lambda \hat{d}$$

F is in newtons and e in metres, so the entries of K are already N/m and Ns/m — exactly what the PID is handed directly.

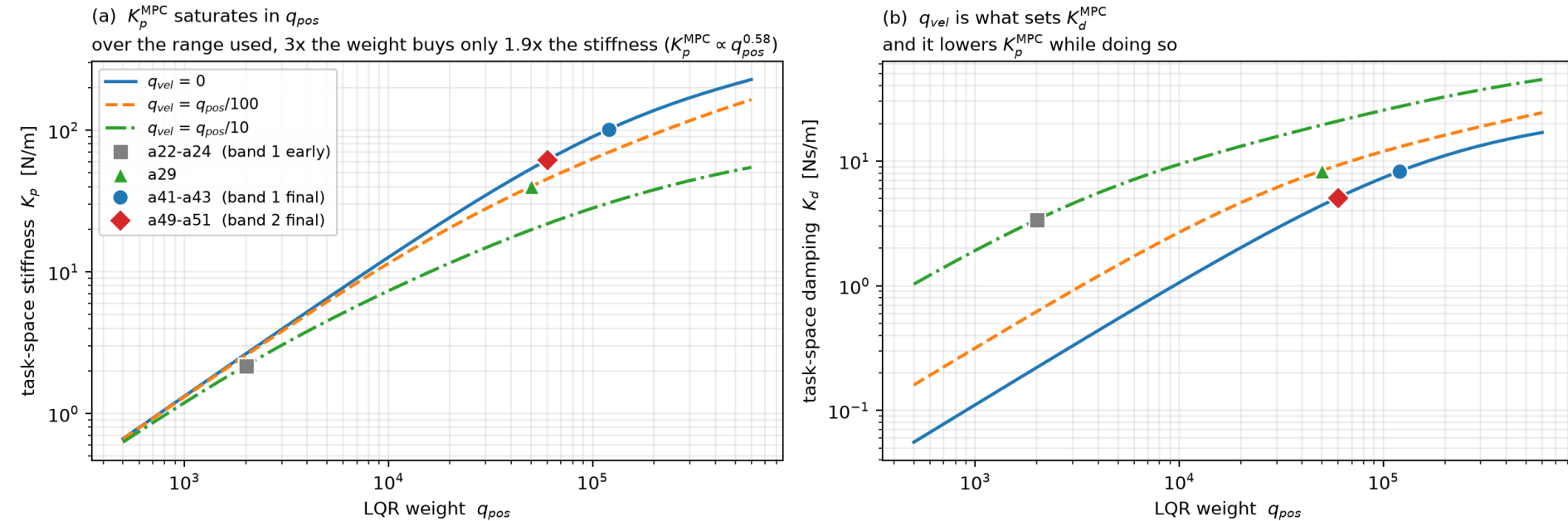
Worked example — the two verified band-1 configurations

	tuned with	becomes
QP-MPC	$q_{pos} = 120000, q_{vel} = 0, r = 10$	$K_p^{\text{MPC}} = 101.4 \text{ N/m}, K_d^{\text{MPC}} = 8.27 \text{ Ns/m}$
PID	—	$K_p^{\text{PID}} = 300 \text{ N/m}, K_d^{\text{PID}} = 6.37 \text{ Ns/m}$

At 1 mm of error the MPC pushes 0.101 N where the PID pushes 0.300 N.

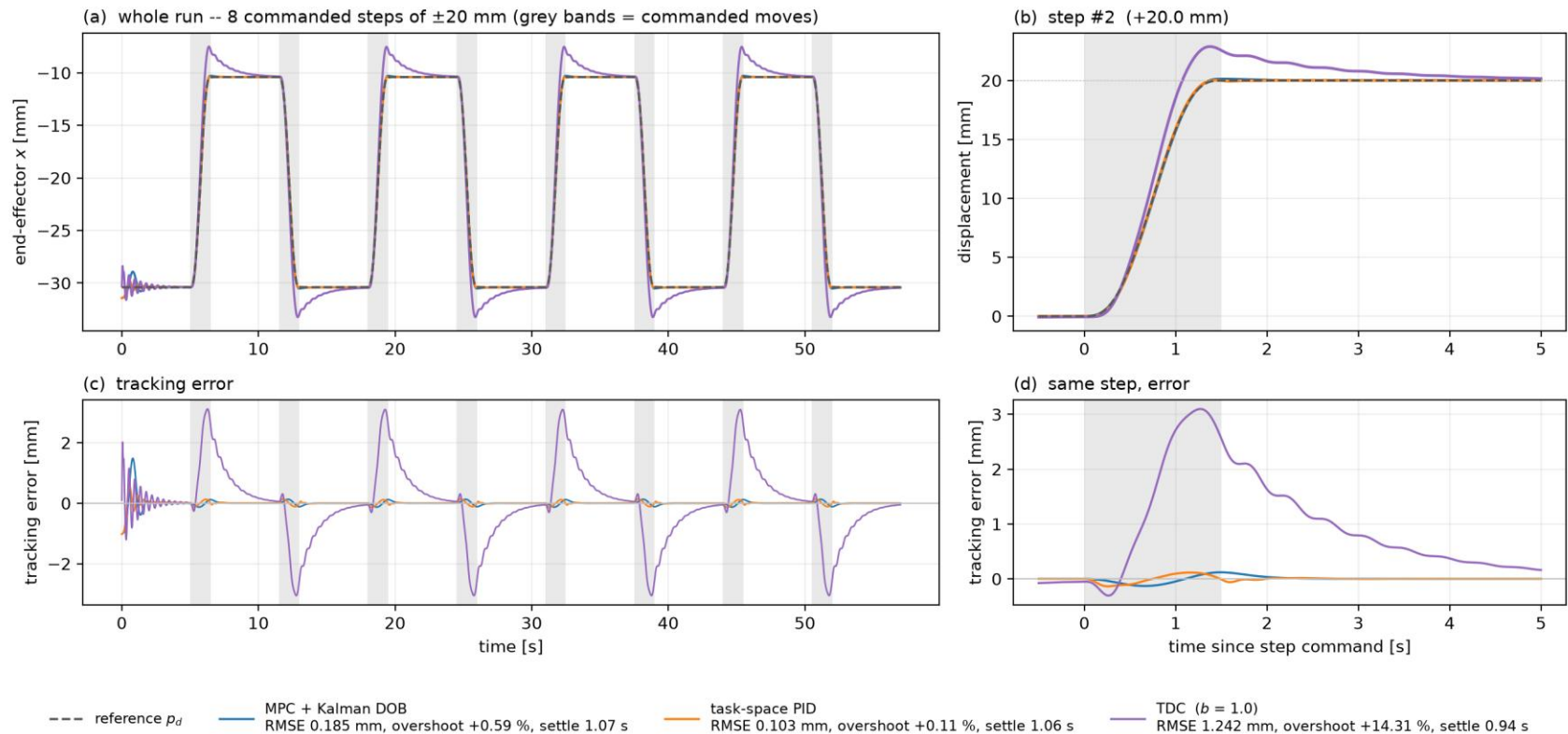
Without the mapping we would be comparing “120000” with “300”.

Mapping the MPC's LQR weights onto the task-space gains the PID uses directly ($r = 10$, $\gamma = 5$, horizon 10)



(a) K_p^{MPC} saturates: over the range used, 3x the weight buys only 1.9x the stiffness. (b) q_{vel} is what sets K_d^{MPC} and it lowers K_p^{MPC} while doing so.

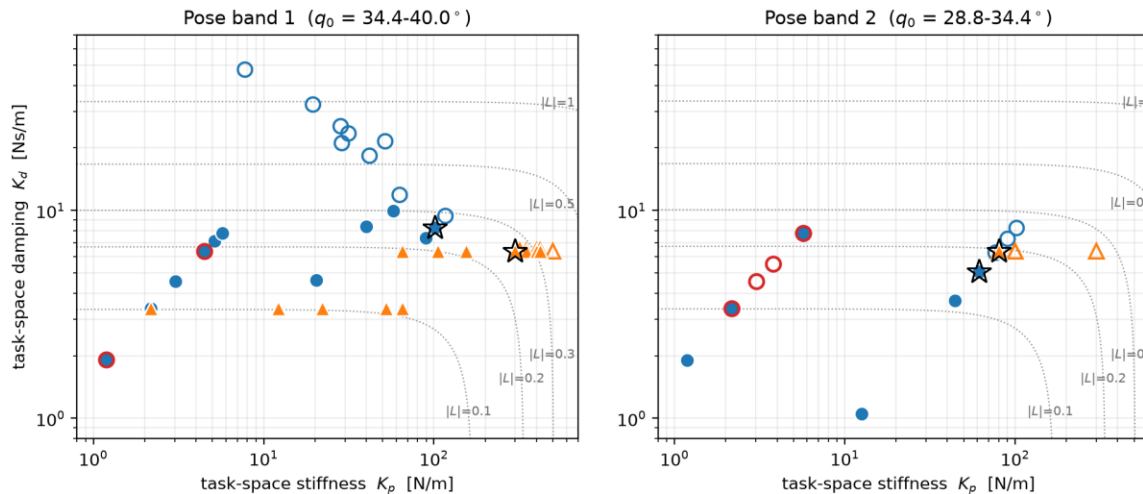
Offen *ii* Simulation, 100 Hz -- MPC, task-space PID and TDC on the identical step reference



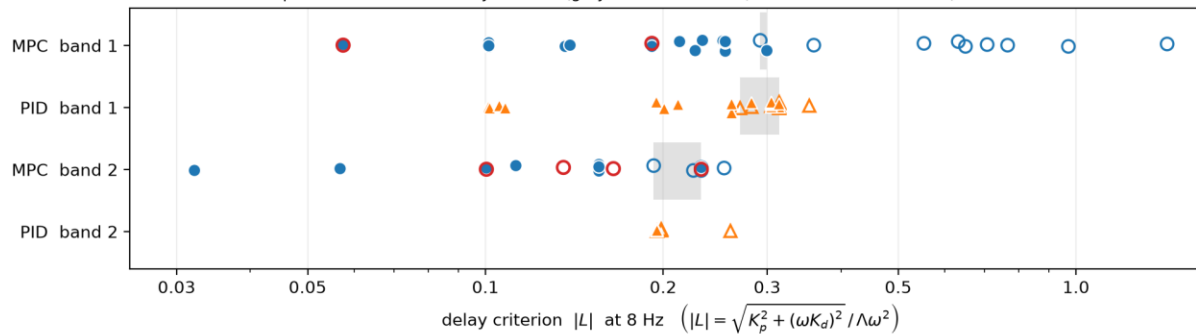
No Coulomb friction, no loop delay $\Rightarrow$ a check that the three laws are correctly implemented. Parameters whose job is to overcome friction cannot be calibrated here.

Stability boundary of both controllers in one task-space gain plane

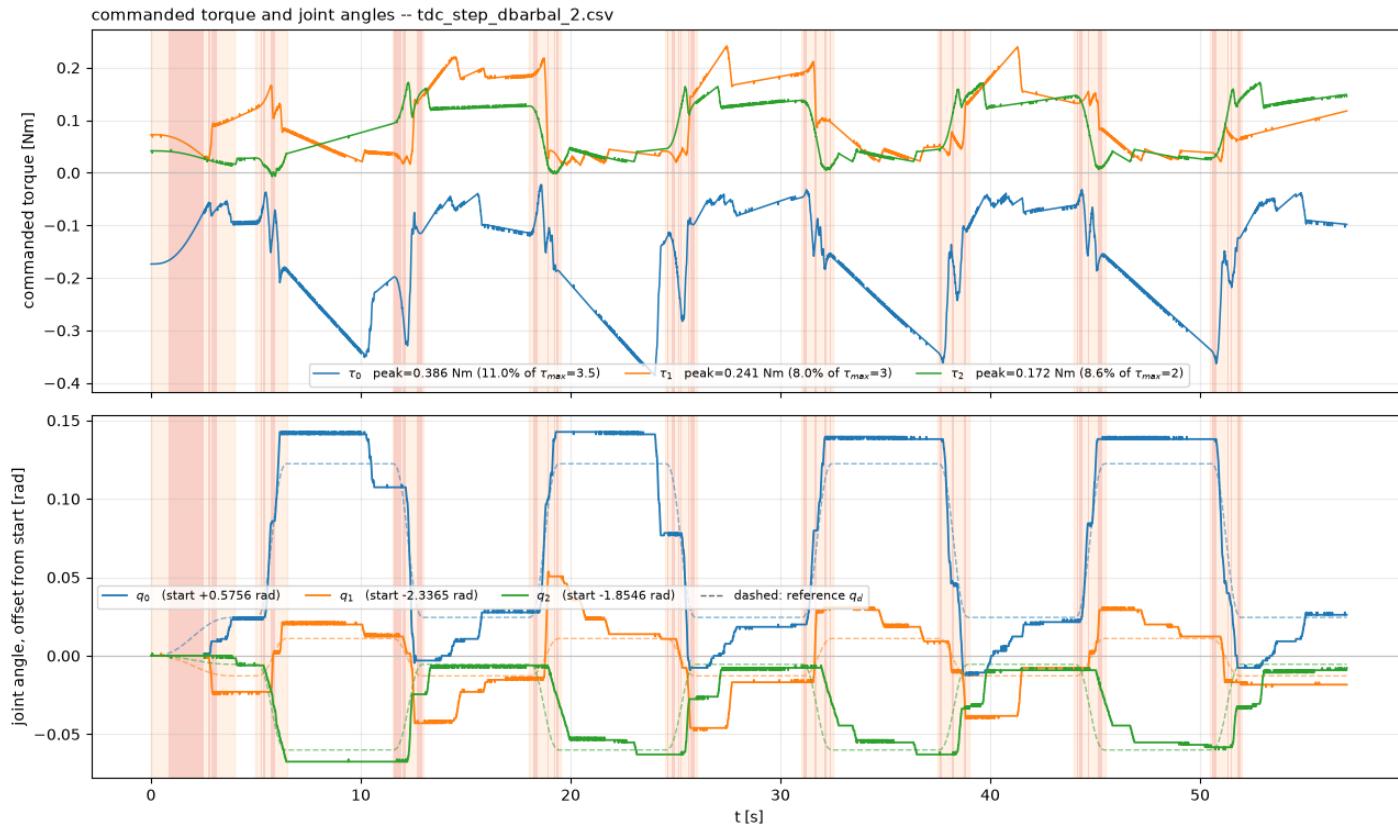
- MPC, completed 57 s ▲ PID, completed 57 s ○ diverged, observer-driven ($q_d \geq 0.05$) constant $|L|$
- MPC, diverged ▲ PID, diverged ☆ final configuration (3 repeats) ■ transition zone (both outcomes)



the same runs collapsed onto the scalar delay criterion (grey = transition zone, both outcomes observed)

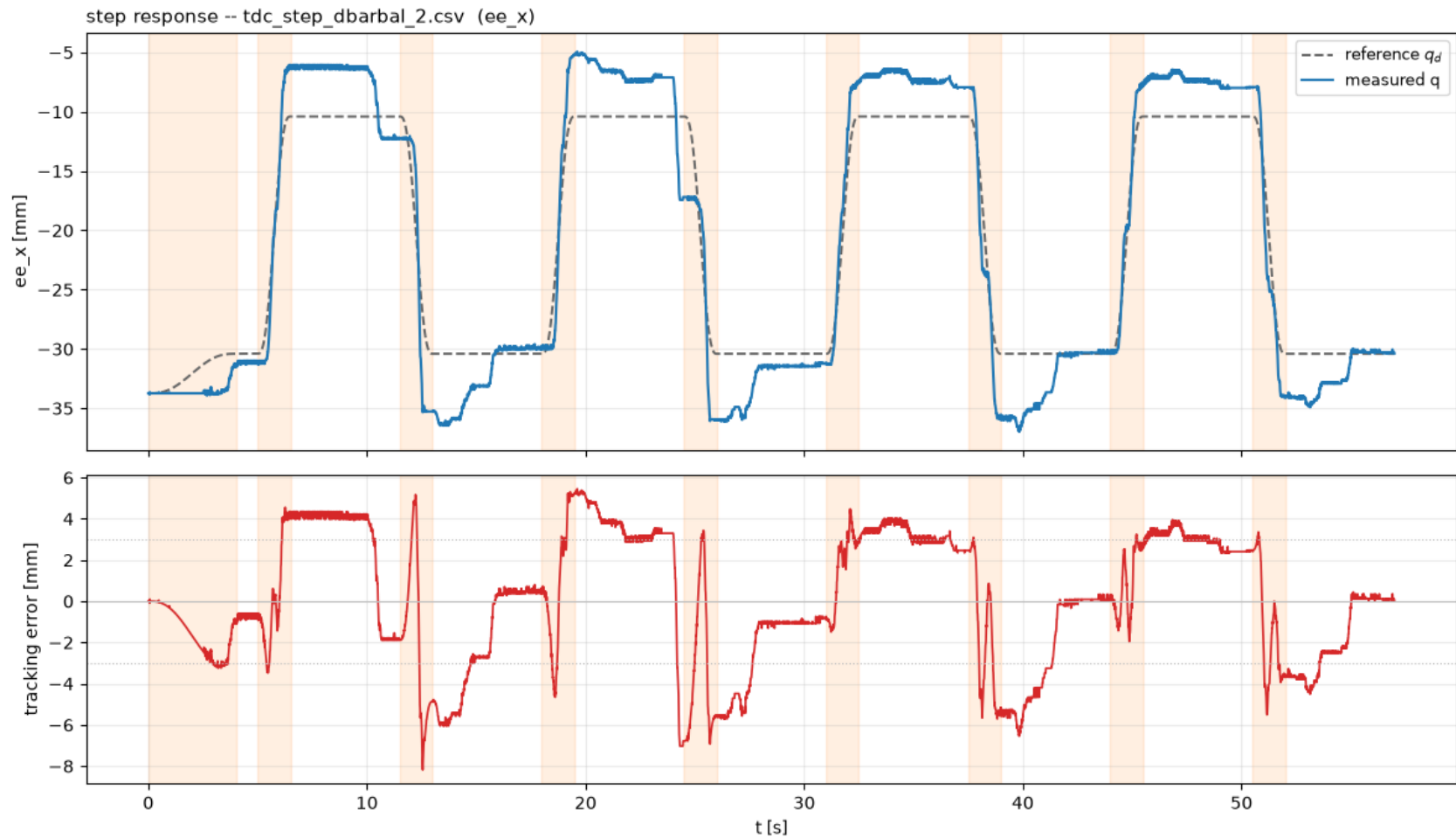


92 runs, 56 completed / 36 diverged. The boundary is nearly horizontal: K_d decides, K_p barely participates. Both controllers obey the same boundary despite a threefold difference in stiffness. Band 2's boundary is lower, which is why both had to be detuned there.



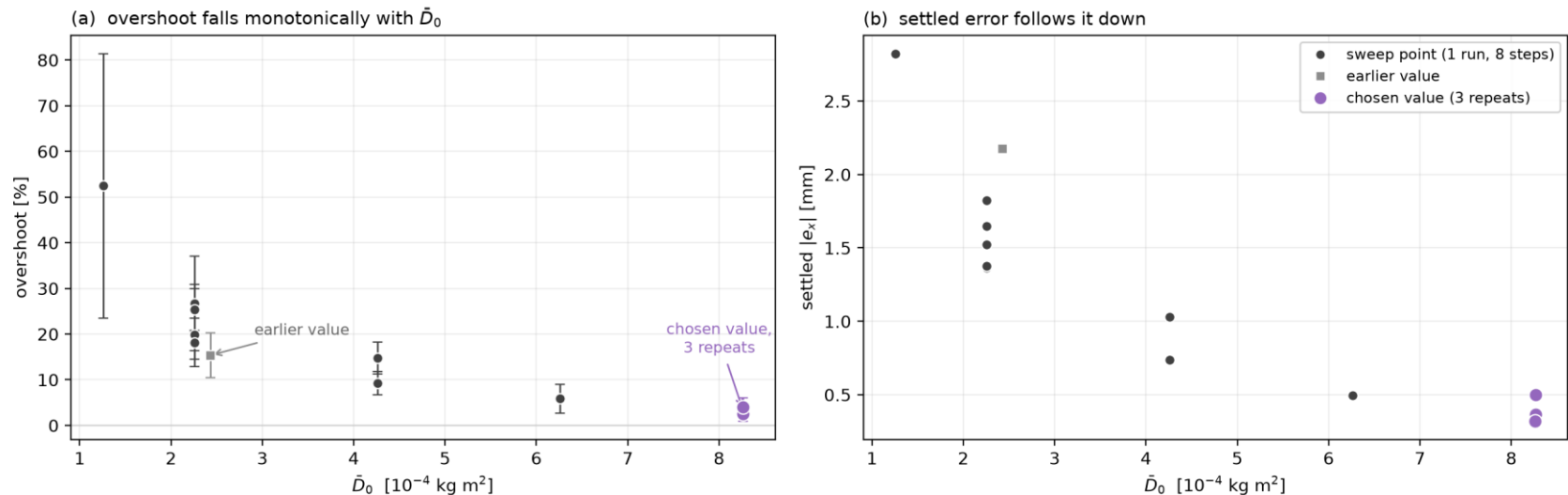
orange = commanded move window red = joint 0 frozen (<0.5 encoder tick) while the reference is still moving, i.e. stuck in stiction; 58% of all commanded-motion samples that percentage rises with the loop rate by construction (a shorter dt crosses a tick less often per sample) -- to compare rates use stall duration in seconds, not this fraction

The controller attempts to reduce steady-state error by increasing the control input; however, once the torque reaches a certain level, the rapid movement of the end-effector instead causes a steady-state error in the opposite direction.



Where TDC is allowed to sit: choosing $\bar{D}$

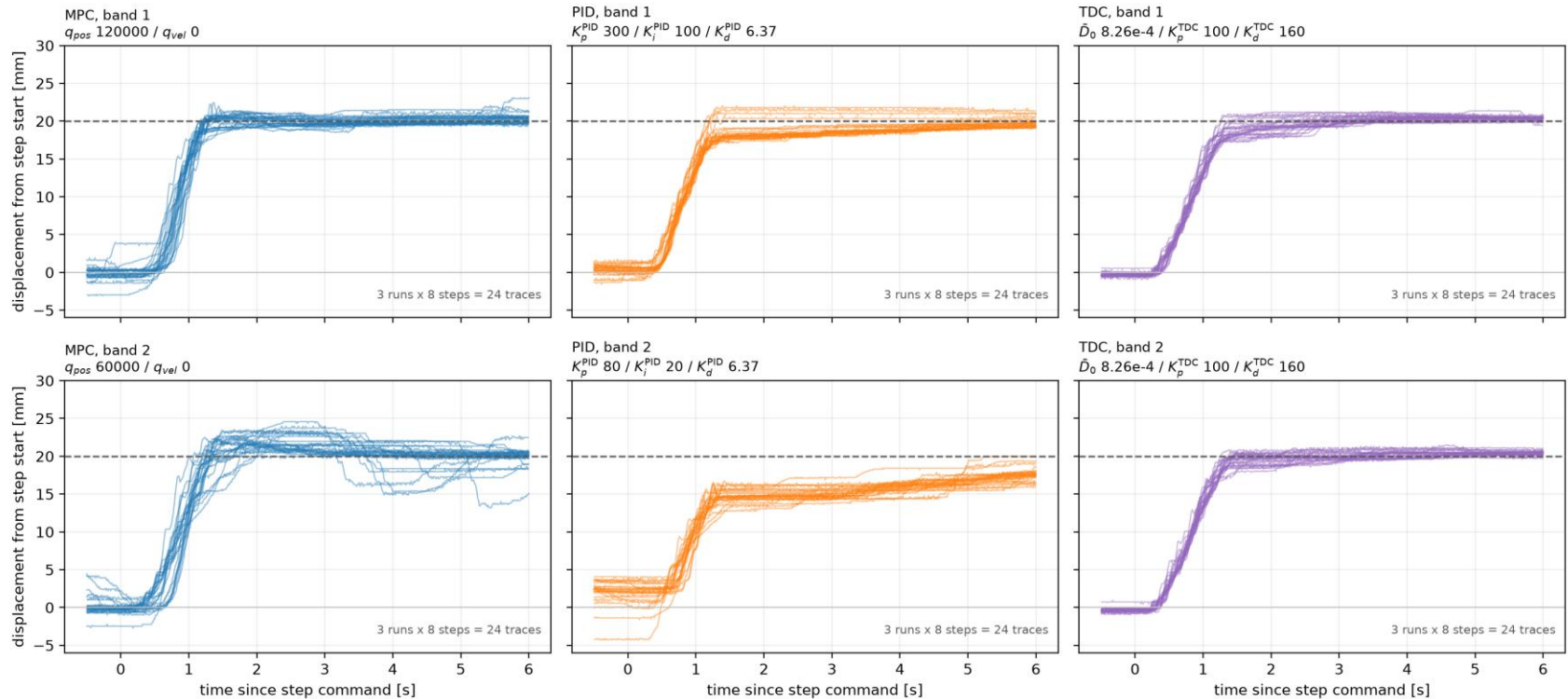
Choosing $\bar{D}_0$: band 1, one run of eight steps per point (K_p^{TDC} 100, K_d^{TDC} 160, $b = 1.0$ throughout)



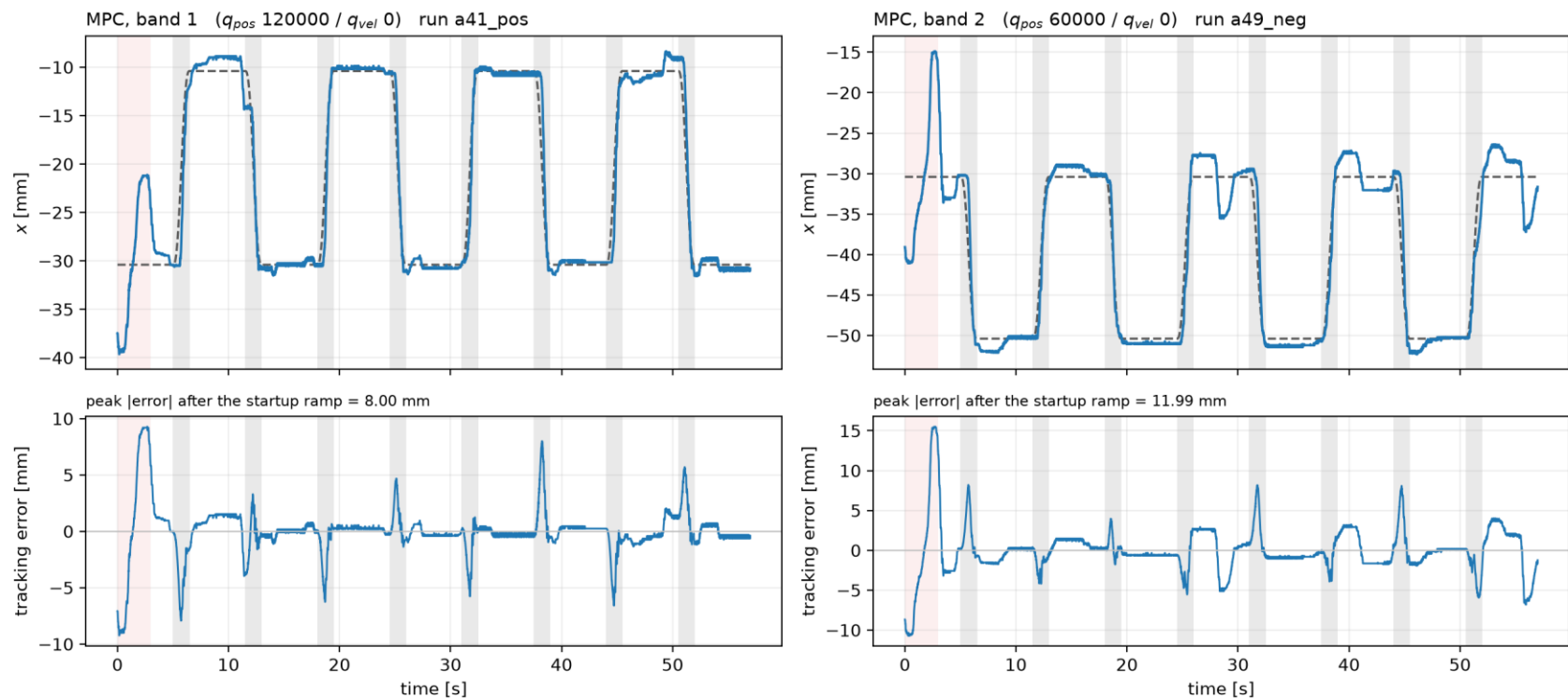
$\bar{D}$ multiplies the whole PD bracket, so it scales loop gain directly. K_p^{TDC}, K_d^{TDC} fixed throughout only $\bar{D}$ moved. Overshoot 52.4% $\rightarrow$ 3.3%, monotone. Every point is one run; the conclusions rest on the 3 repeats at the chosen value. Swept on band 1 only, then transferred to band 2 unchanged.

Repeatability – 6 cells × 3 runs × 8 steps

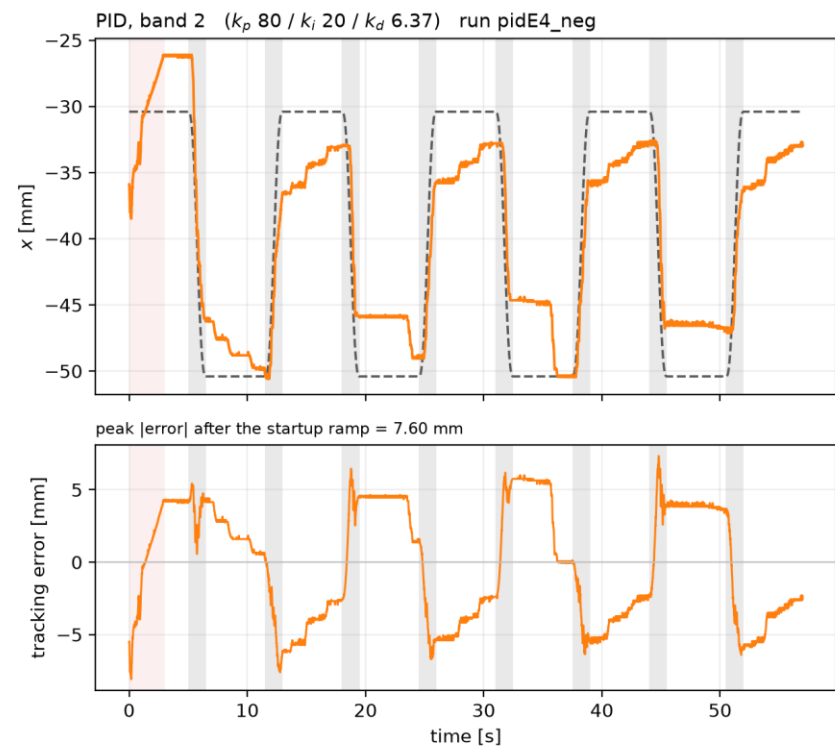
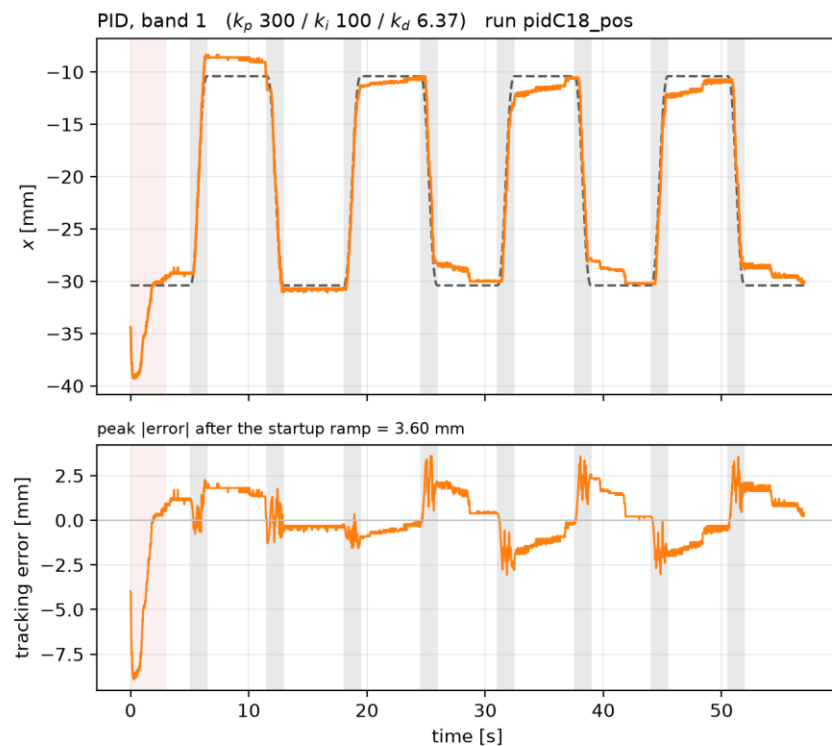
Repeatability: all steps of all repeats, sign-normalised (dashed line = commanded 20 mm)



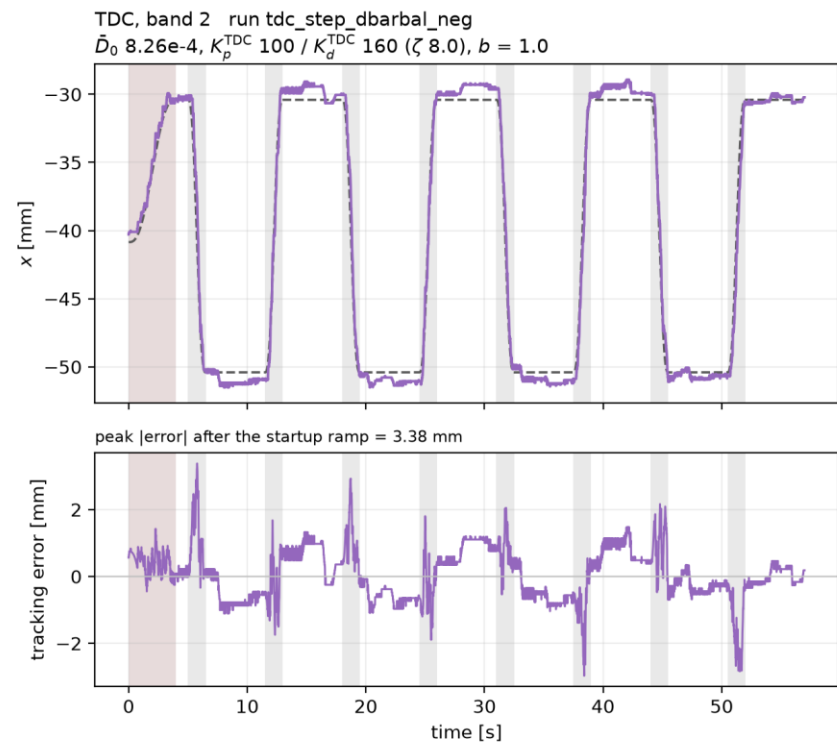
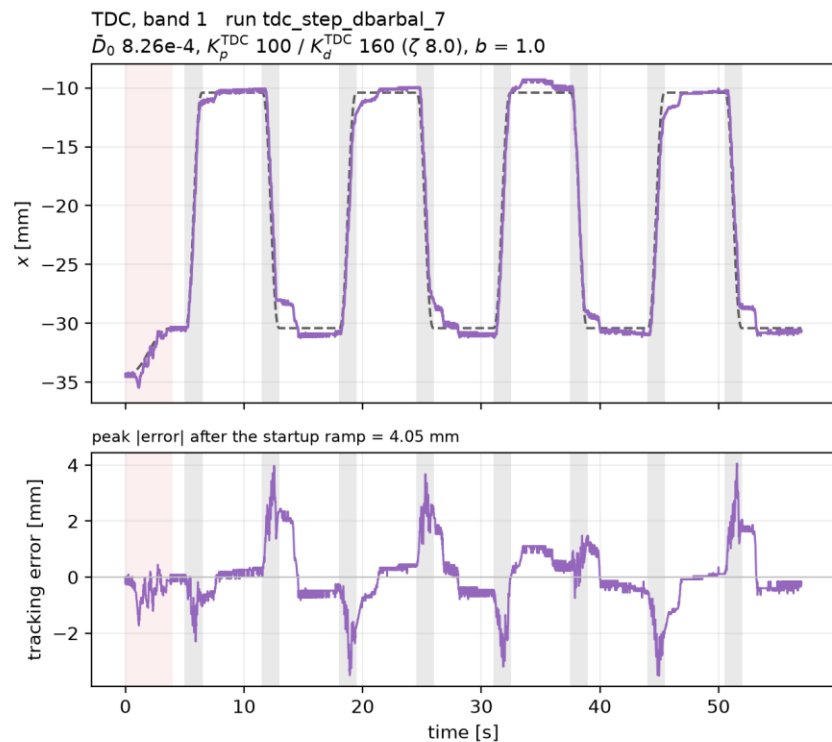
Hardware overview -- MPC + Kalman DOB (dashed = reference, solid = measured; red band = startup ramp, grey bands = commanded moves)



Hardware overview -- task-space PID (dashed = reference, solid = measured; red band = startup ramp, grey bands = commanded moves)

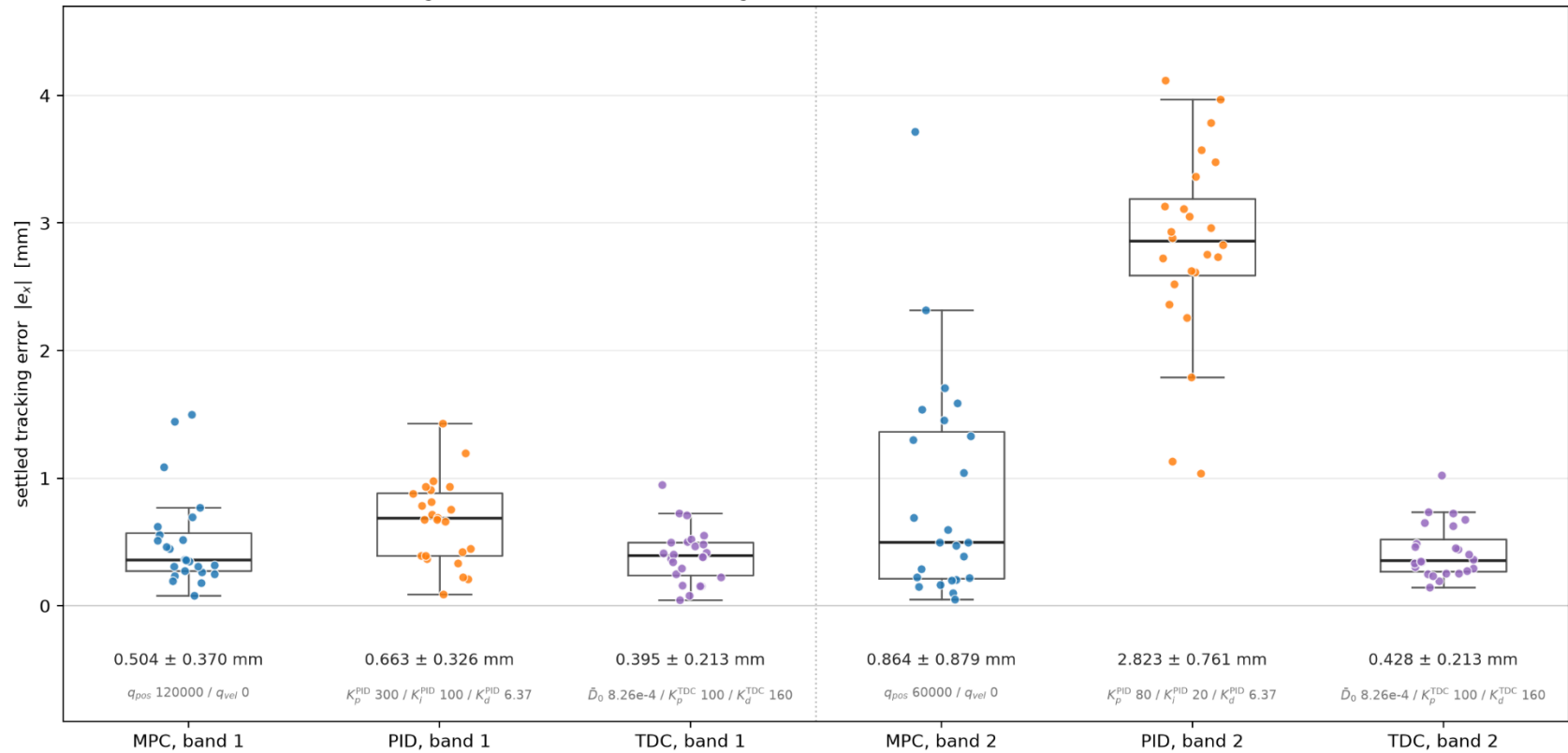


Hardware overview -- Time-Delay Control @ 100 Hz (dashed = reference, solid = measured; red band = startup ramp, grey bands = commanded moves)



Settled tracking error, 6 cells x 3 runs x 8 steps = 144 steps.

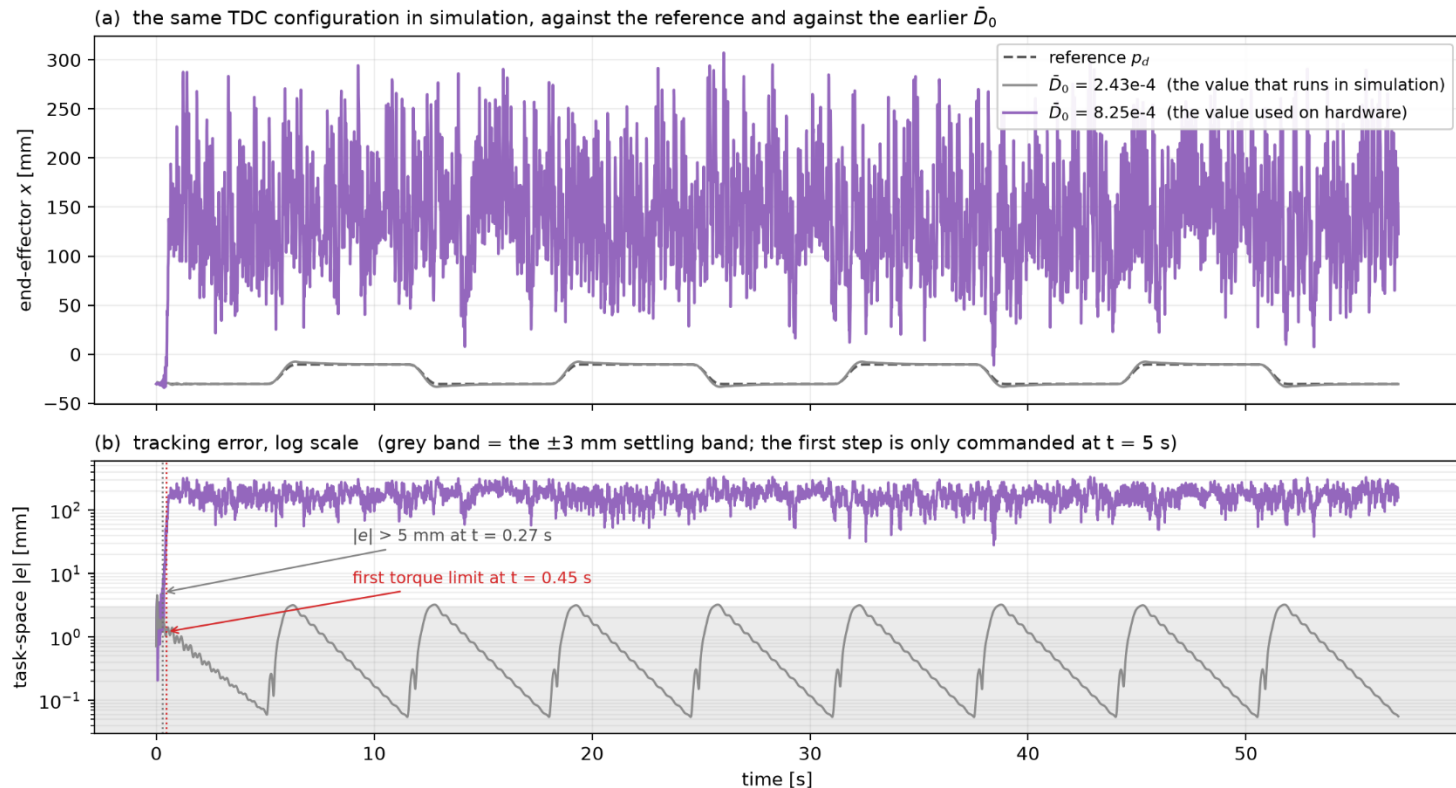
Each cell is annotated with its own gains -- for MPC and PID, the highest that held 3/3 runs in that band.



Controller	Band	$ L $	$ e_x $ [mm]	rise [s]	overshoot [%]	settling [s]
MPC	1	0.255	0.504 ± 0.362	0.544 ± 0.155	$+5.68 \pm 3.50$	1.536
PID	1	0.261	0.663 ± 0.319	1.095 ± 0.471	$+0.28 \pm 3.58$	1.187
TDC	1	—	0.395 ± 0.209	0.985 ± 0.327	$+3.31 \pm 1.75$	1.185
MPC	2	0.156	0.864 ± 0.861	0.697 ± 0.312	$+11.31 \pm 5.27$	2.163
PID	2	0.195	2.823 ± 0.745	2.145 ± 2.849	-10.47 ± 4.73	5.415
TDC	2	—	0.428 ± 0.209	0.827 ± 0.094	$+3.95 \pm 1.49$	1.192

TDC vs PID significant in both bands; TDC vs MPC significant in band 2 ($p = 0.026$), not distinguishable in band 1.

The hardware-verified TDC configuration, run in simulation: it leaves the hold pose before the first step is commanded



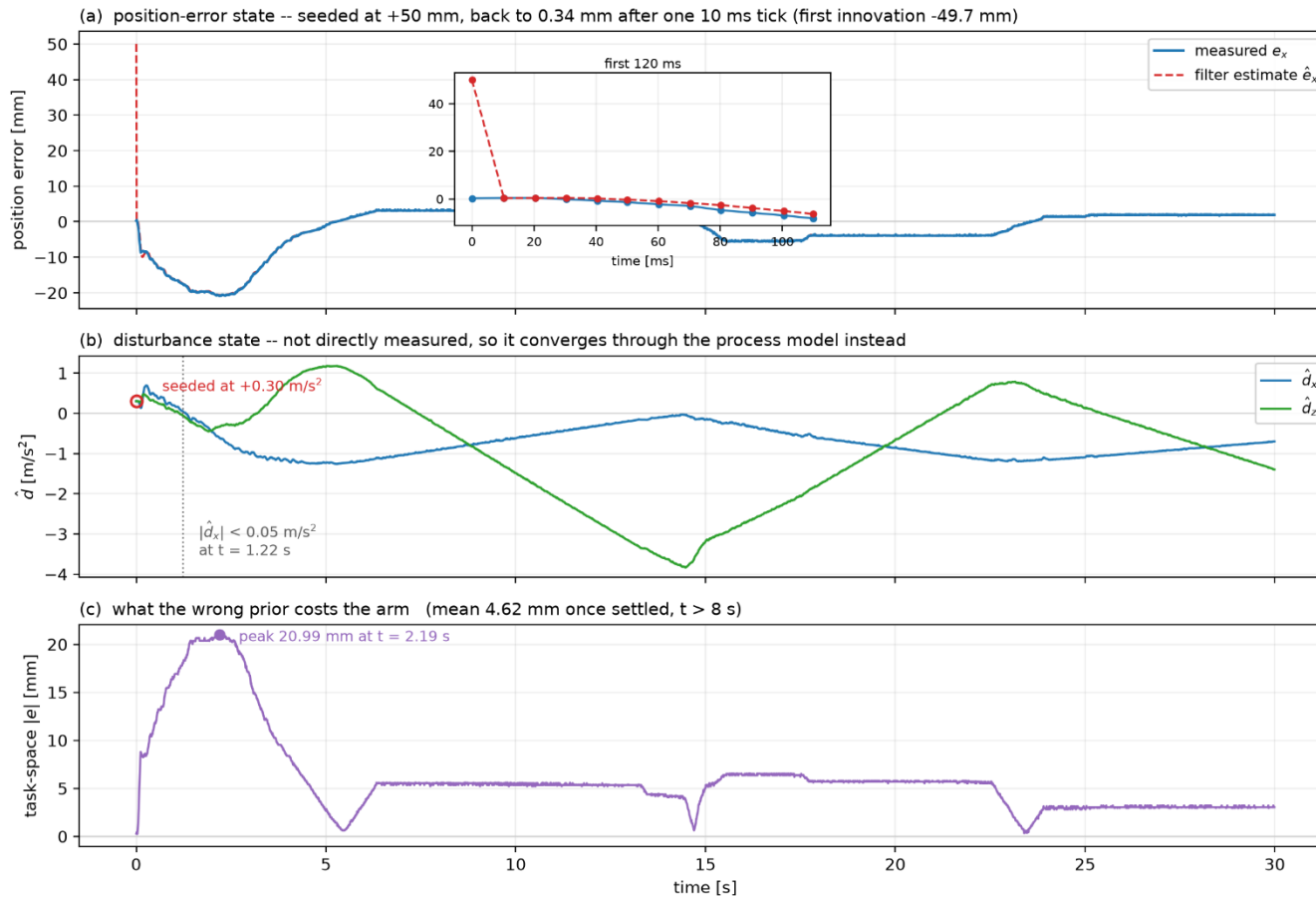
The same configuration that wins on hardware leaves the hold pose in 0.45 s in simulation, before the first step is even commanded. $\bar{D}$ raised 3.40× raises joint 1's effective gain 3.40×; on hardware that buys stiction authority, in a frictionless simulator it is simply too large gain.

Controller	mean [ms]	p99 [ms]	max [ms]	p99 as % of period
MPC (a41_pos)	8.17	10.49	13.64	105 %
Task-space PID (pidC18_pos)	6.73	8.87	37.57	89 %
TDC (dbarbal_7)	2.71	5.30	7.60	53 %

The MPC exceeds the 10 ms period, it is compute-bound here and cannot be run faster

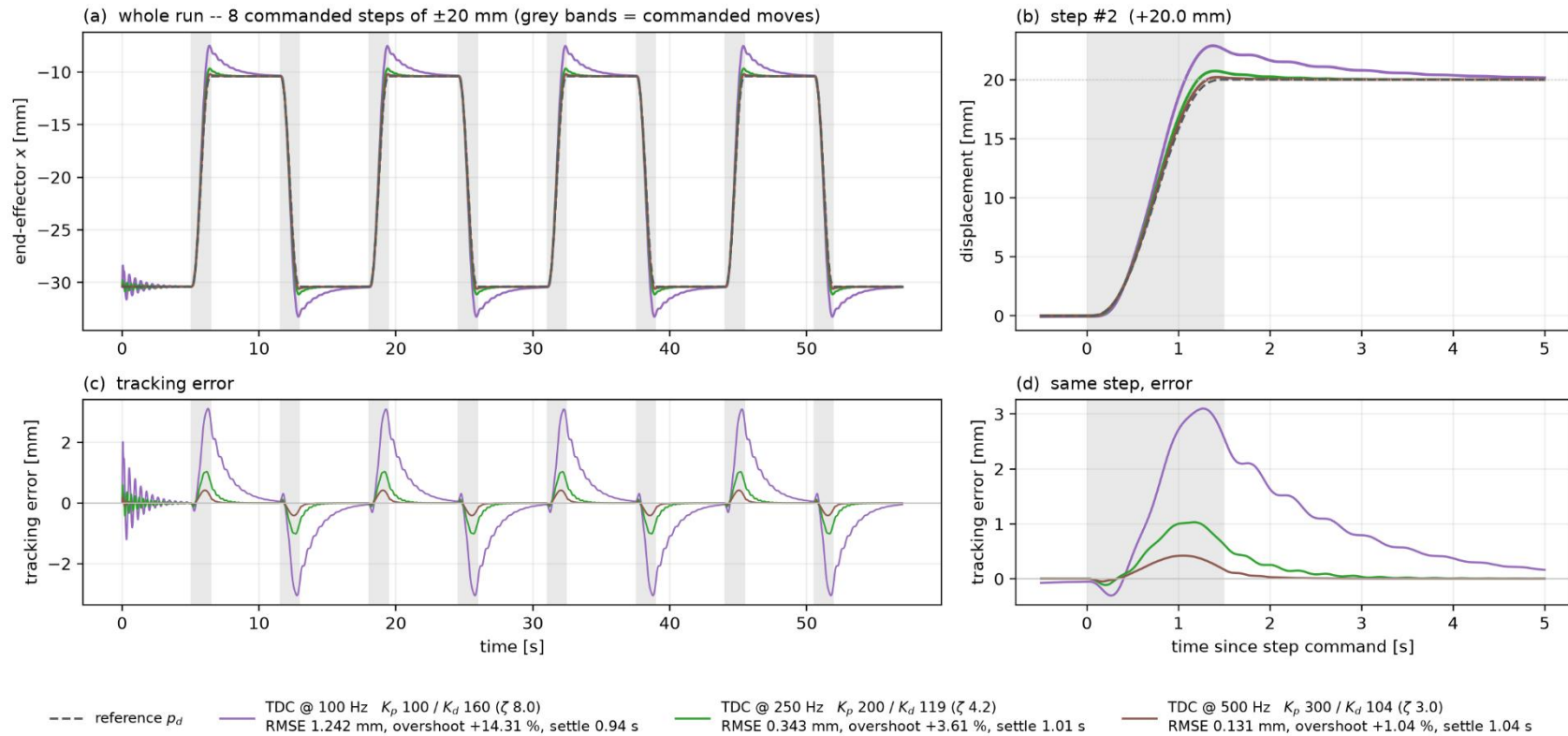
- The arm is friction-dominated: the measured task-space breakaway force is 0.94 N, and the TDC configuration that produced one of the best hardware result of this study diverges within 0.45 s in a frictionless simulation.
- Neither of the two properties that motivate task-space control, commandable compliance and redundancy resolution, was exercised by this step-tracking protocol, so these results say nothing about which formulation is preferable in general.

Fig. 3.2 Observer convergence from a deliberately wrong initial state (hardware, hold trajectory)

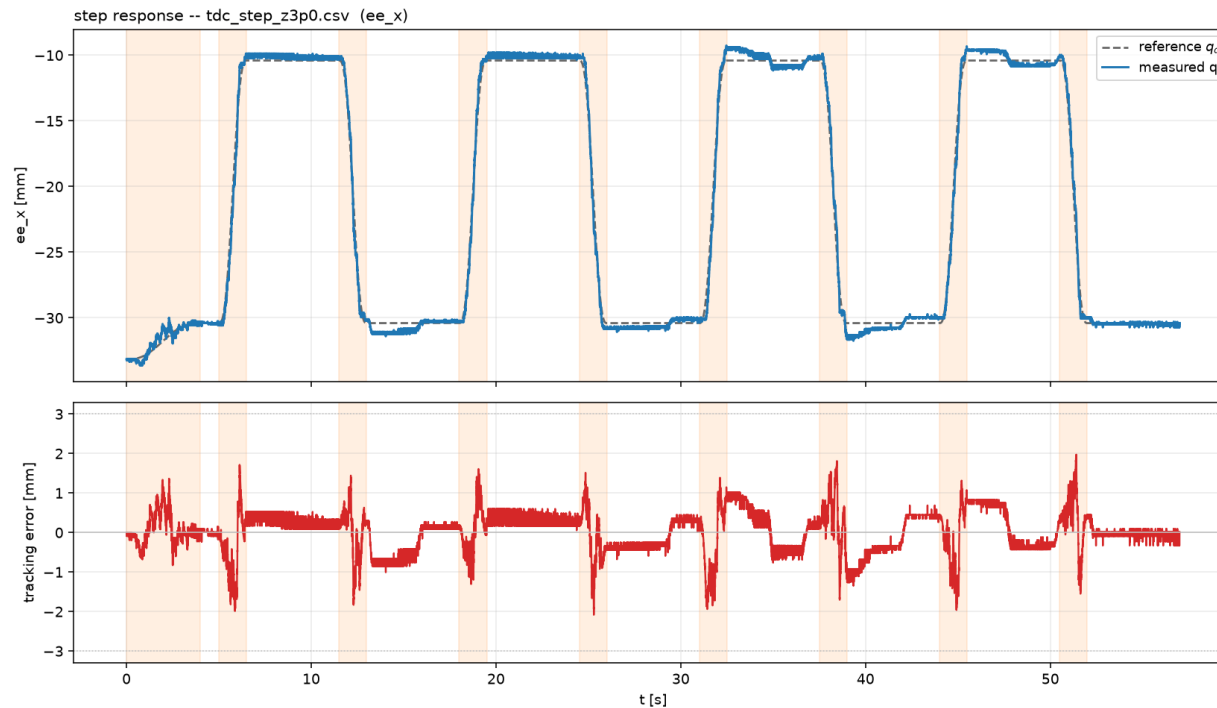


Backup: TDC at three loop rates (simulation)

Fig. 5.2 TDC in simulation at three loop rates -- each at its own lowest-RMSE tuning (gains do not transfer across dt)



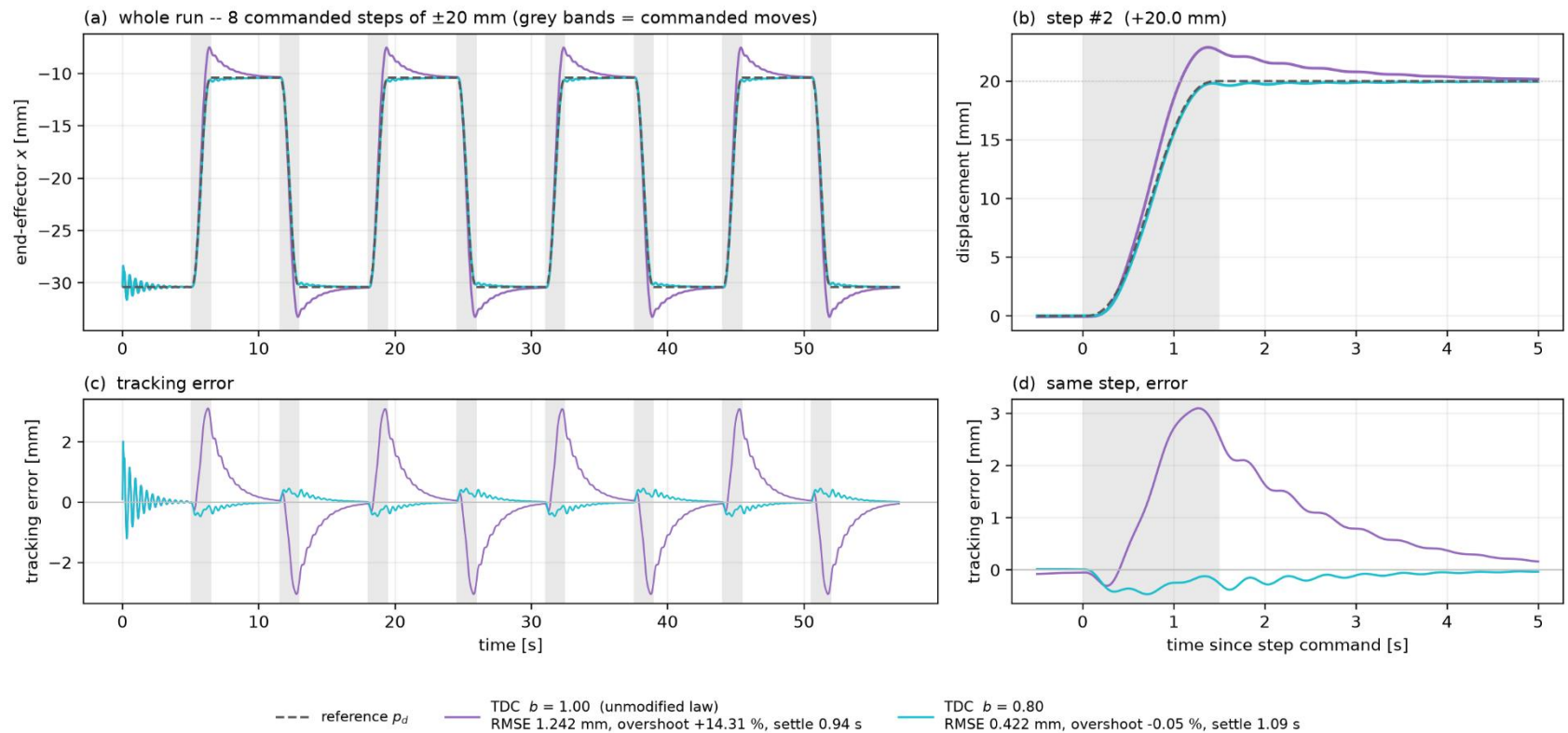
Backup: TDC at 500 Hz on hardware



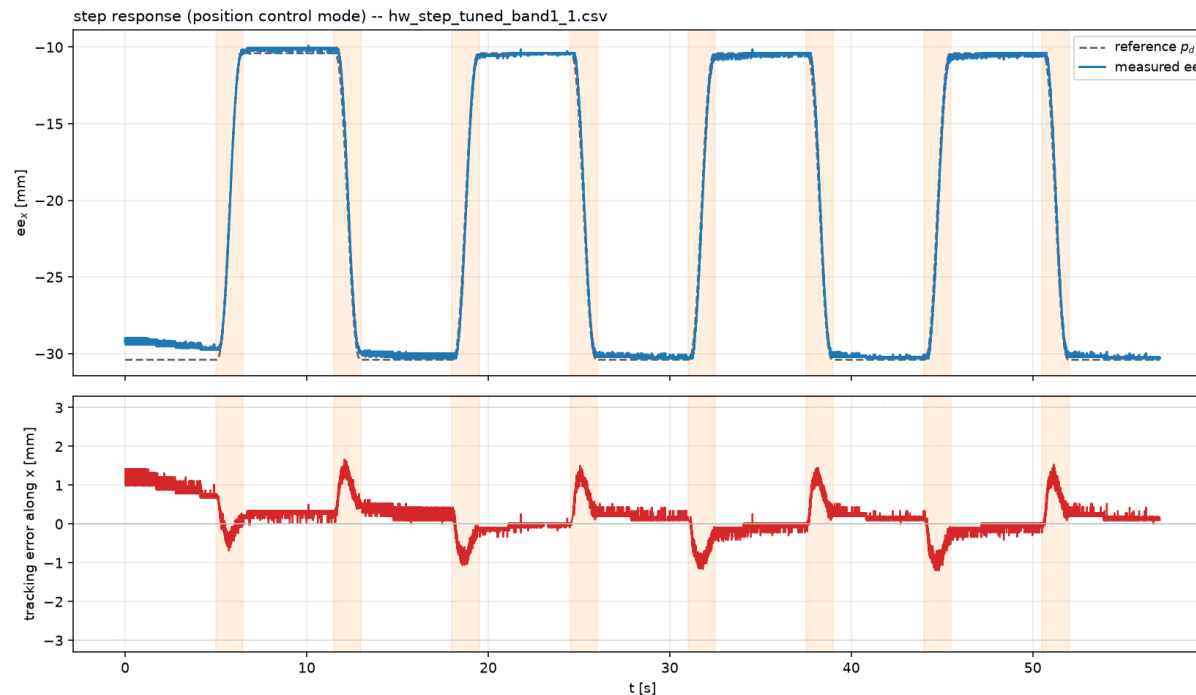
Same protocol as the 100 Hz result, $n = 24$ per band. $|e_x|$: $0.395 \rightarrow 0.222$ mm (band 1), $0.428 \rightarrow 0.207$ mm (band 2); rise time and settling both improve too

Backup: the $b < 1$ route

$$\dot{q}_{\text{meas}} = \frac{q(t) - q(t-1)}{dt}, \quad \dot{q}_d = \dot{e}_q + \dot{q}_{\text{meas}}, \quad \dot{e}_q \leftarrow b \dot{q}_d - \dot{q}_{\text{meas}}$$

Fig. 5.3 TDC at 100 Hz -- derivative setpoint weighting b against the unmodified law ($b = 1$), simulation only

Backup: native Position Control Mode



Default gains ($D=0$, $I=0$, $P=800$) leave a gravity-dependent steady-state error (0.955 mm); raising the servo's own I gain alone — no code change on either side — takes it to 0.079 mm.